

WHERE WE LEFT OFF

five asks • the test each was given • today

THE ASK	TEST SET IN ADVANCE		TODAY
OVS RECOVERY	detect + recompute + install = 51.75 s	ANSWERED	it balances • 87% is detection
FASTER DETECTION	false positives, not detection time	ANSWERED	3.9× faster • idle false positives 0
FAST BUILD DEFAULT	L0–L4 suite passes on the fast build	PREREQ	binaries separable • suite not run
SFLOW JITTER	one fabric • swap kernel, then rate	ANSWERED	inherited • stops paying at 1-in-32
TRUNCATE / MERGE	either moves the ceiling, or cost ≠ per byte	IN PART	truncate: no • merge: wall unmeasured

THE FAILOVER BUDGET

128 hosts · one inter-switch link down · n = 10

QUESTION

Where do the 51.75 s go?

the test was that the terms add up, not that a cause is found

RESULT

detect 44.9 s · debounce 3.0 · recompute 0.25

SO

87% is waiting to notice · both suspects < 5%

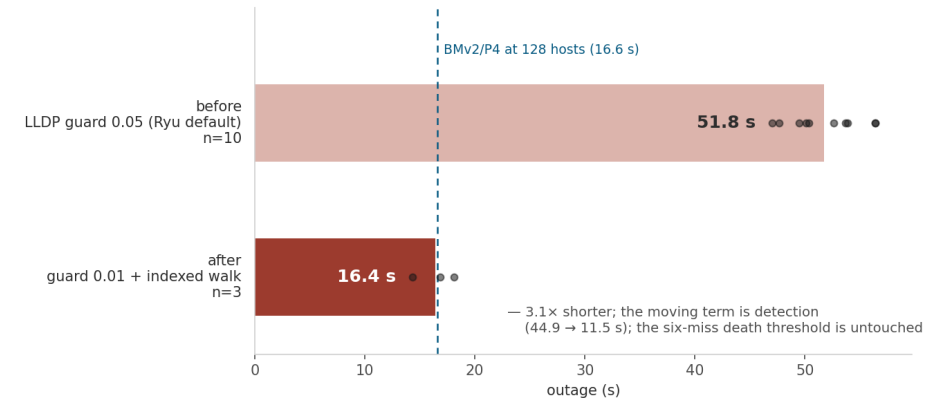
FIX

one constant · probe 0.05 → 0.01 · 51.8 → 16.4 s

LIMIT

idle false positives only · threshold untouched

The same link failure, before and after the fix



THE 45-POINT THREAD

one boot · CPU trace and kernel log from the same run

THREAD	1/1024	1/32	WHAT IT IS
calFlowPathByQueried	46.31%	46.84%	1 kHz path recompute
run	3.98%	13.86%	the sFlow ingest
testCalAvgFlowSendingRates...	0.00%	0.00%	never runs
purgeIdleFlows	0.00%	0.00%	never runs

- FINDING

32× the samples · +0.53 points · sampling is not the cost
- IT IS

a path recompute at 1 kHz · inherited · present at 28b8b13
- INGEST

3.98 → 13.86% · ×3.48 with the rate · this is the cost
- NAMED HOW

log and trace from one boot · tid offset moves per build

PRE-REGISTERED, THEN WITHDRAWN

six cells · two levels · one control

QUESTION	Does the same sample count mean the same precision?
CONFIRMED	pairing held · 0.84% and 0.44% · six cells healthy
ABOUT TO CLAIM	cells not equivalent ⇒ the matrix cannot be skipped
RESTED ON	every pair clearing 0.1 · one cleared it by 0.040
CONTROL	same parameters ×3 · spread moved 0.136 · range 0.240
WITHDRAWN	the threshold cannot separate cell from cell
NOT TAKEN	widening to 0.24 rescues it — registration forbids that

The reversal was already in a draft of this deck. The control is why it is not in this one.

WHERE IT STANDS

settled · open · what the next report answers

SETTLED

Failover budget	balances · detection 87%
Detection	3.9× · false positives 0
A sample	206 μs · switches flat
Precision	stops paying at 1-in-32
Truncation	−5.6× bytes · ceiling unmoved
Concurrency	reads high · four rounds

OPEN

Boot convergence	verified at N = 3 only
Fast build	L0–L4 not run
Merge	wired · wall unmeasured
Ceiling mechanism	receive-side · inferred
Scale × concurrency	14% between generations
Loaded false positives	idle only
Residual	3.6 s · unexplained

Marked items are what the next report answers.